Staphylococcus Discitis: A Comprehensive Disease Characterization

Infectious Disease | MONDO:8000001 | Category: Infectious Disease

Summary

Staphylococcus discitis is an acquired bacterial infection of the intervertebral disc and adjacent vertebral endplates/bodies, caused predominantly by *Staphylococcus aureus* (MSSA or MRSA). It is not a genetic or heritable disorder; rather, it is one manifestation along a contiguous spectrum of spine infection that includes discitis, vertebral osteomyelitis, and epidural abscess. A landmark pediatric study concluded that non-tuberculous spondylodiscitis "is truly osteomyelitis of the spine" (PMID: 11961447), and contemporary reviews define vertebral osteomyelitis as "a general term for a heterogenous group of spine infections involving the vertebral bone, intervertebral discs, and facet joints" (PMID: 40204566). Because it is infectious rather than Mendelian, entire categories in a classic disease template — OMIM inheritance, causal germline variants, penetrance, anticipation, founder effects — are **not applicable**; only polygenic host susceptibility to invasive *S. aureus* is relevant.

The disease chiefly affects **older adults** (mean age ~71 years, ~61% male in a Japanese nationwide cohort of 71,134 patients) with an incidence of roughly **3 cases per 100,000 per year**, plus a distinct **toddler subgroup** (mean age ~4.5 years) in which *S. aureus* and *Kingella kingae* dominate. The **lumbar spine** is most commonly affected (63%), and the classic presentation is **back pain** (~79%) with **fever** (~72%) and elevated inflammatory markers. Diagnosis rests on **contrast-enhanced MRI** (imaging gold standard) supplemented by **18F-FDG PET/CT** and confirmed by **blood culture or image-guided biopsy**. Bone destruction is driven mechanistically by *S. aureus*–induced osteoblast and immune signaling: surface protein A binds osteoblast **TNFR-1**, activating **JNK/TLR2** signaling and driving **RANKL**-mediated osteoclastic bone resorption.

Management combines **pathogen-directed antibiotics** — a **6-week course was proven non-inferior to 12 weeks** in a 359-patient randomized controlled trial — with **surgery** reserved for neurologic deficit, spinal instability, drainable abscess, or medical failure. **Timely surgery**

(≤14 days) improved survival in staphylococcal cohorts. Prognosis is generally favorable but not benign: **mortality is ~3–12%**, **relapse ~6–17%**, and **residual disability ~20–30%**, with radicular nerve damage and epidural/paravertebral abscess being frequent complications. Novel salvage options (dalbavancin, bacteriophage therapy) are emerging for refractory or multidrug-resistant disease.

Key Findings

Finding 1 — *S. aureus* is the leading causative pathogen of pyogenic spondylodiscitis

Across geographically diverse cohorts, *Staphylococcus aureus* is consistently the single most common organism. In a 14-year retrospective study of 98 infectious spondylodiscitis patients from Trieste, *S. aureus* accounted for 34 cases — **64% of pyogenic spondylodiscitis** (PMID: 41023355): *"Staphylococcus aureus was the most common pathogen, with 34 cases, representing 64% of pyogenic spondylodiscitis."* A Danish cohort of 106 pyogenic vertebral osteomyelitis patients found **39% *S. aureus*** (PMID: 34091001): *"39% were infected with Staphylococcus aureus and 9% with Escherichia coli."* In the largest dataset — a Japanese nationwide database of 71,134 pyogenic spondylodiscitis patients — **MRSA alone comprised 11.9%** (PMID: 39716636), and in surgically treated US spinal infections MRSA reached 43%. This establishes *S. aureus* — spanning both methicillin-susceptible (MSSA) and methicillin-resistant (MRSA) strains — as the defining pathogen of the disease entity.

Finding 2 — Epidemiology: ~3/100,000/yr incidence, elderly male predominance, lumbar predilection

The incidence of infectious spondylodiscitis is reported at **3 cases per 100,000 inhabitants per year** (PMID: 41023355). The Japanese nationwide cohort documented a **mean age of 71.05 years and 60.9% male** (PMID: 39716636): *"The average age of patients was 71.05 years, and 60.9% of patients were male."* Anatomically, disease favors the **lumbar spine (63%)**, followed by thoracic (25%) and cervical (12%) (PMID: 41023355). Presenting symptoms are dominated by back pain (~79%) and fever (~72%). The suggested anatomical ontology anchor is **UBERON:0002103 (intervertebral disc)**, with lumbar region UBERON:0002792.

Feature	Value	Source
Incidence	3 / 100,000 / yr	PMID: 41023355
Mean age	71.05 yr	PMID: 39716636
Male proportion	60.9%	PMID: 39716636
Lumbar / thoracic / cervical	63% / 25% / 12%	PMID: 41023355

Finding 3 — Pediatric discitis: young children, *S. aureus* and *Kingella kingae*, good outcome

A distinct pediatric form affects toddlers. In 42 children with non-tuberculous spondylodiscitis (mean age 4 years 6 months), bacteria were isolated in 22/35 samples: **55% *S. aureus*, 27% *Kingella kingae***, and one *Coxiella burnetii* (PMID: 11961447): *"Bacteria were isolated in 22 of the 35 samples taken (55% Staphylococcus aureus, 27% Kingella kingae; Coxiella burnetii in one sample)."* Diagnosis was frequently delayed (mean 42 days). Functional outcome is generally good with appropriate treatment, though long-term disc fibrosis and occasional vertebral fusion develop. Crucially, the authors framed discitis as a spinal osteomyelitis — *"nontuberculous spondylodiscitis is truly osteomyelitis of the spine"* — reinforcing the unified nosology (Finding 13).

Finding 4 — MRSA risk factors: smoking, IV drug use, thoracic involvement, more complications

MRSA disease carries a distinct risk profile. In a US cohort of 205 surgical spinal infections, MRSA (43%) versus non-MRSA patients had significantly higher **current smoking (57% vs 40%, p=.021), IV drug use (49% vs 33%, p=.021), thoracic involvement (43% vs 28%, p=.039), and epidural abscess (65% vs 45%, p=.005)** (PMID: 40653018). More generally, predictors of inadequate treatment response include **previous spinal surgery (aOR 19.29; 95% CI 2.20–169.08) and severe sepsis (aOR 4.59; 95% CI 1.28–15.41)** (PMID: 34091001). Diabetes mellitus, prior spinal instrumentation, bacteremia, and immunosuppression are recurrent risk themes across cohorts.

Finding 5 — Six weeks of antibiotics is non-inferior to twelve weeks (RCT)

The pivotal therapeutic evidence is an open-label, non-inferiority randomized controlled trial (NCT00764114) of **359 patients across 71 French centers** with microbiologically confirmed pyogenic vertebral osteomyelitis, randomized to 6 versus 12 weeks of antibiotics (PMID: 25468170): *"We aimed to establish whether 6 weeks of antibiotic treatment is non-inferior to 12 weeks in patients with pyogenic vertebral osteomyelitis."* Using a masked committee endpoint of cure at 1 year and a non-inferiority margin of 10%, **6 weeks proved non-inferior to 12 weeks**. This finding is now embedded in AO Spine recommendations (PMID: 39852953) as a conditional recommendation for a 6-week course in confirmed cases.

Finding 6 — Mechanism: protein A → TNFR-1 → JNK/TLR2 → RANKL-driven bone destruction

The molecular pathophysiology of staphylococcal bone destruction has been dissected in vitro. *S. aureus* surface **protein A (SpA) binds directly to pre-osteoblastic cells via TNF receptor-1 (TNFR-1)** (PMID: 26846887): *"the S. aureus protein SpA binds directly to pre-osteoblastic cells via tumor necrosis factor receptor-1 (TNFR-1)."* Infection decreased osteoblast viability and increased apoptosis, TLR2 expression, and TNF-α/IL-6 secretion, with activation of the **JNK pathway**. Critically, *"Knockdown of TNFR1 not only inhibited the JNK pathway but also reduced TLR2 protein and RANKL levels in S. aureus-infected cells"* — linking the receptor interaction to **RANKL-driven osteoclastic bone resorption**, the effector of the "widespread bone loss and destruction" characteristic of osteomyelitis.

Suggested ontology terms: GO:0045672 (regulation of osteoclast differentiation), GO:0007254 (JNK cascade), GO:0002224 (toll-like receptor signaling pathway), GO:0006954 (inflammatory response); CL:0000062 (osteoblast), CL:0000092 (osteoclast).

Finding 7 — Diagnosis: MRI is the gold standard; 18F-FDG PET/CT is a complementary adjunct

Contrast-enhanced MRI is the imaging cornerstone, with PET/CT adding complementary value. A prospective head-to-head comparison (n=26 haematogenous spondylodiscitis, 18 confirmed) found **18F-FDG PET/CT sensitivity 83%, specificity 88%, PPV 94%, NPV 70%; MRI sensitivity 94%, specificity 38%, PPV 77%, NPV 75%** (PMID: 25186431): *"The sensitivity, specificity, and positive and negative predictive value were 83%, 88%, 94% and 70% for (18)F-FDG PET/CT, and 94%, 38%, 77% and 75% for MRI, respectively."* A retrospective cohort (n=191) showed combining modalities raised sensitivity: *"sensitivity was 56% for FDG-PET/CT and 62% for MRI,*

increasing to 74% when both modalities were combined (p = 0.002)" (PMID: 42232200). MRI accuracy is time-dependent: *"Diagnostic accuracy was 58%, when MRI was performed within 2 weeks after the start of symptoms and improved to 82%, when performed more than 2 weeks after onset of symptoms"* (PMID: 27317050). Microbiological confirmation via blood culture or image-guided biopsy remains essential; open/intraoperative sampling yields more positives than percutaneous (RR 1.65, 95% CI 1.34–2.03) (PMID: 41519476).

Modality	Sensitivity	Specificity	PPV	NPV
18F-FDG PET/CT	83%	88%	94%	70%
MRI	94%	38%	77%	75%

Source: PMID: 25186431

Finding 8 — Prognosis: mortality 3–12%, frequent abscess/neurologic complications, relapse 6–17%

Outcomes are generally favorable but complications are common. The Trieste cohort (n=98) reported **radicular nerve damage (51%), epidural/paravertebral abscesses (45%), endocarditis (14%), need for surgery (8%), persistent neurological sequelae (3%), and infection-related mortality 6%** (PMID: 41023355). The Danish PVO cohort reported **13% relapse within 2 years and 1-year crude mortality of 12%** (PMID: 34091001). A surgical *Staphylococcus* cohort reported **1-year overall survival 76% and recurrence-free survival 58%** (PMID: 42314843): *"The 1-year OS and RFS were 76% and 58%, respectively."* A systematic review of culture-negative versus culture-positive disease reported recurrence 6–17% and residual disability 20–30% (PMID: 42262500). Concurrent infections (especially cardiac, CNS, pneumonia) roughly quadruple early mortality (3.3% vs 0.8%) (PMID: 37148755).

Finding 9 — Animal models: the New Zealand White rabbit is the established model

Staphylococcal discitis has a well-validated animal model. Percutaneous fluoroscopy-guided inoculation of *S. aureus* into rabbit lumbar discs reproducibly produces disease. Bierry et al. (12 NZW rabbits) found **disc-space infections in all animals**, with MRI detecting abnormalities 5–10 days post-inoculation (PMID: 18485785): *"Disc space infections were present in all animals. MRI detected vertebral and discal abnormalities as soon as, respectively, 5 and 10 days after inoculation."* The model has been used for antibiotic-penetration studies —

clindamycin penetrated the nucleus pulposus at 11.9–20% of serum versus cephazolin's 0.68–0.77% (PMID: 24043368) — and comparative pathogen studies showing *S. aureus* produces true discitis with higher death rates (2/3) than *P. acnes* (PMID: 26925420). Histology recapitulates disc-space narrowing, endplate erosion/hyperostosis, epidural abscess, and paraplegia. **Model organism:** *Oryctolagus cuniculus* (NCBI:txid9986).

Finding 10 — Host genetic susceptibility to invasive *S. aureus* is polygenic (iron regulation)

There is **no monogenic cause** of staphylococcal discitis — it is an acquired infectious disease. Host susceptibility to invasive *S. aureus* is multifactorial/polygenic. A multi-site African GWAS (n=3,928 children plus replication) found the **CHCHD7/SDR16C5 variant rs73596248**, which increases hepcidin, *"is associated with reduced risk of Klebsiella pneumoniae and Staphylococcus aureus bacteremia"* (PMID: 41946712), linking iron regulation to host–pathogen susceptibility. Classic monogenic immunodeficiencies that predispose to staphylococcal disease include Job/hyper-IgE syndrome (**STAT3, DOCK8**), chronic granulomatous disease (**CYBB**/NADPH oxidase), and **IRAK4/MyD88** deficiency (impaired TLR/IL-1 signaling) — but these are rare and not specific to discitis.

Finding 11 — Salvage/novel therapeutics: dalbavancin and bacteriophage therapy

For refractory or difficult-to-treat staphylococcal spinal infection, novel options are emerging. **Dalbavancin**, a long-acting lipoglycopeptide, was studied in 83 bone/joint infections (48.2% vertebral osteomyelitis/spondylodiscitis) and *"offers a simplified alternative"* to prolonged IV therapy (PMID: 41305347). A striking case of **systemic bacteriophage therapy** in a 60-year-old woman with chronic implant-associated MRSA vertebral osteomyelitis and antibiotic hypersensitivity used 35 IV doses of 10^9 PFU anti-staphylococcal phage (PMID: 41478341): *"Systemic bacteriophage therapy was well tolerated, with no adverse events observed. The treatment resulted in sustained clinical improvement and normalization of inflammatory markers,"* with no relapse 6 months after stopping all antibiotics. First-line therapy remains anti-staphylococcal penicillins/cefazolin (MSSA) and vancomycin/daptomycin/linezolid (MRSA), with rifampicin added for implant/biofilm-associated disease.

Finding 12 — Treatment strategy: antibiotics ± surgery, with defined surgical indications

Management combines pathogen-directed antibiotics (per IDSA 2015 guidelines, PMID: 26229122) with surgery for specific indications. Predictors of needing surgery (n=80) include **pyogenic etiology (81.6% vs 38.1%, p<0.001), thoracic localization (36.8% vs 16.7%, p=0.04), epidural abscess (52.6% vs 11.9%, p<0.001), psoas abscess (21.1% vs 4.8%, p<0.001), and Gram-positive coccal infection (50% vs 7.1%, p<0.001)** (PMID: 40762806). **Timely surgery (≤14 days)** improved outcomes in staphylococcal spondylodiscitis: *"timely surgery (≤14 days) were associated with improved outcomes, demonstrating significantly better OS (HR: 0.18, p=.003; HR: 0.22, p=.019) and RFS"* (PMID: 42314843). Conservative treatment remains viable for neurologically intact patients (PMID: 39119368). Standard surgical indications: neurologic deficit, spinal instability/deformity, large/drainable abscess, medical failure, and need for microbiological diagnosis.

Finding 13 — Nosology: staphylococcal discitis is part of the spondylodiscitis/vertebral osteomyelitis spectrum

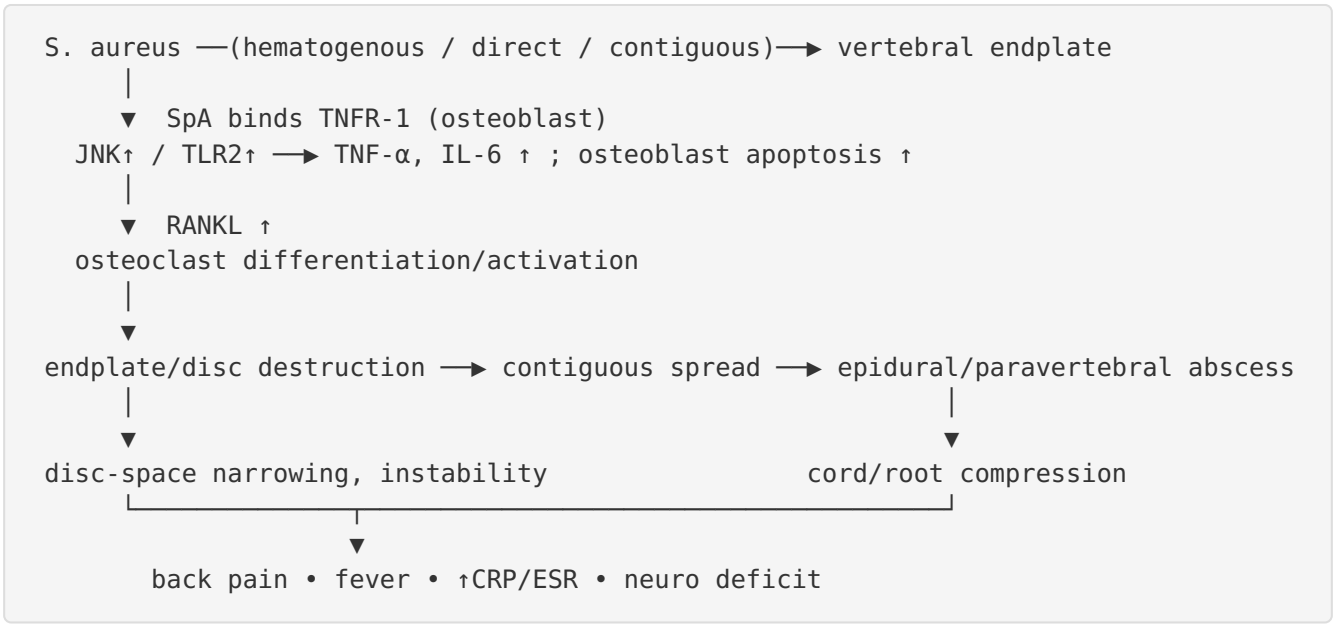
Discitis, vertebral osteomyelitis, and epidural abscess form a **contiguous spine-infection spectrum**. A 2025 review defines vertebral osteomyelitis as *"a general term for a heterogenous group of spine infections involving the vertebral bone, intervertebral discs, and facet joints"* that *"spread contiguously to adjacent paraspinal structures, including paravertebral muscles and the epidural space,"* with *"nonspecific and indolent symptoms, which can delay diagnosis"* (PMID: 40204566). The pediatric conclusion that discitis *"is truly osteomyelitis of the spine"* (PMID: 11961447) reinforces this unified view.

Suggested identifiers: MONDO:8000001; ICD-10 **M46.4** (discitis, unspecified) + **B95.6** (*S. aureus* as cause of disease); ICD-11 **FA84** (spondylopathies / infection of intervertebral disc); MeSH **Discitis (D021782)**, **Staphylococcal Infections (D013203)**; primary anatomy **UBERON:0002103** (intervertebral disc).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Bacterial seeding of the disc/endplate.** *S. aureus* reaches the vertebral endplate via **(a) hematogenous seeding** through end-arteriolar vascular arcades (the dominant route in adults and children), **(b) direct inoculation** from spinal surgery/instrumentation, or **(c) contiguous spread** from adjacent infection — *leads to* colonization of the richly vascularized subchondral endplate. *(Well demonstrated across human cohorts and rabbit models — Findings 1, 9.)*
2. **Adherence and receptor engagement.** Bacterial surface protein A (SpA) **binds osteoblast TNFR-1** — *results in* activation of intracellular stress signaling. *(In vitro demonstrated — Finding 6.)*
3. **Signal transduction.** TNFR-1 engagement **activates the JNK pathway and upregulates TLR2** — *leads to* secretion of TNF-α and IL-6 and osteoblast apoptosis. *(In vitro demonstrated — Finding 6.)*
4. **Osteoclast activation.** The same signaling **increases RANKL** on osteoblasts/stromal cells — *results in* RANK-mediated osteoclast differentiation and activation. *(In vitro demonstrated; TNFR1 knockdown reduces RANKL — Finding 6.)*
5. **Bone and disc destruction.** Osteoclast-mediated resorption plus direct bacterial/inflammatory injury **erode the endplate and cartilage**, and the avascular nucleus pulposus (poor antibiotic penetration) becomes a nidus — *leads to* disc-space narrowing, endplate erosion, and hyperostosis. *(Inferred from combined human histology and rabbit models — Findings 9, 6.)*
6. **Local spread / complications.** Infection **extends contiguously** into paravertebral muscle, psoas, and the epidural space — *results in* abscess formation, and where the epidural abscess or vertebral collapse compresses neural elements, *leads to* radiculopathy, cord compression, and paraplegia. *(Human cohorts — Findings 8, 12.)*
7. **Clinical manifestation.** The net result is **back pain, fever, elevated CRP/ESR**, and — in advanced cases — neurologic deficit and spinal instability. *(Human cohorts — Findings 2, 8.)*



Upstream vs downstream: The bacterial receptor engagement (steps 2–3) is upstream; RANKL-driven resorption and abscess/neurologic complications (steps 4–6) are downstream effectors. **Cell types involved:** osteoblasts (CL:0000062), osteoclasts (CL:0000092), neutrophils/macrophages (inflammatory infiltrate), and nucleus pulposus cells. **Biological processes:** inflammatory response (GO:0006954), osteoclast differentiation (GO:0030316), toll-like receptor signaling (GO:0002224), JNK cascade (GO:0007254).

Section-by-Section Synthesis

1. Disease Information

Staphylococcal discitis is an infectious inflammation of the intervertebral disc and adjacent endplates caused predominantly by *S. aureus*. Synonyms/related terms: infective/pyogenic spondylodiscitis, septic discitis, vertebral osteomyelitis (when bone-predominant), spinal osteomyelitis. Information derives from **aggregated disease-level resources** (cohorts, registries, guidelines), not single-patient EHR. Identifiers: MONDO:8000001; MeSH Discitis D021782; ICD-10 M46.4 + B95.6; ICD-11 FA84. No OMIM/Orphanet entry applies because it is not a Mendelian disorder.

2. Etiology

Primary cause: infectious — *S. aureus* (MSSA/MRSA) predominates (Finding 1). Routes: hematogenous, direct inoculation, contiguous. **Risk factors:** advanced age, male sex, diabetes mellitus, IV drug use, prior spinal surgery/instrumentation, bacteremia, immunosuppression, smoking (Findings 2, 4). **Genetic risk** is polygenic host susceptibility to invasive *S. aureus* (Finding 10); no causal disease gene. **Protective factors:** avoidance/treatment of bacteremia sources, aseptic surgical technique, treatment of asymptomatic bacteriuria before urologic instrumentation (PMID: 42249268); the hepcidin-raising CHCHD7/SDR16C5 allele is genetically protective against *S. aureus* bacteremia (Finding 10).

3. Phenotypes

Phenotype	Type	Frequency	HPO
Back pain	Symptom	~79%	HP:0003418 (back pain)
Fever	Sign	~72%	HP:0001945
Elevated CRP/ESR	Lab	very common	HP:0011227 (elevated CRP)
Radiculopathy / radicular nerve damage	Sign	~51%	HP:0500152 (radicular pain)
Epidural/paravertebral abscess	Manifestation	~45%	HP:0012062 (epidural abscess)
Neurologic deficit / paraplegia	Sign	variable	HP:0002385 (paraplegia)

Onset is adult (geriatric peak) or toddler; severity variable; progression subacute/insidious. QoL impact: substantial pain, disability, prolonged hospitalization, residual disability 20–30%.

4. Genetic/Molecular Information

Not applicable as a heritable disorder. No causal genes, pathogenic germline variants, ACMG classifications, allele frequencies, modifier genes, or chromosomal abnormalities. Host susceptibility is polygenic (Finding 10). Relevant *bacterial* molecular determinants: protein A (*spa*), *sae* regulon (master regulator SaeS, PMID: 41172037), Panton-Valentine leukocidin, biofilm genes.

5. Environmental Information

Infectious agent: **Staphylococcus aureus** (NCBI:txid1280); pediatric co-agent **Kingella kingae** (NCBI:txid504). Lifestyle: IV drug use and smoking increase MRSA risk (Finding 4). Iatrogenic exposures: spinal surgery, epidural procedures, urologic instrumentation.

6. Mechanism / Pathophysiology

See the ordered causal chain above (protein A → TNFR-1 → JNK/TLR2 → RANKL → osteoclastic destruction → contiguous spread).

7. Anatomical Structures Affected

Primary: intervertebral disc (UBERON:0002103) and vertebral endplate/body; lumbar > thoracic > cervical. **Secondary:** epidural space, paravertebral/psoas muscle, spinal cord/nerve roots; occasionally heart valves (endocarditis, 14%). Body systems: musculoskeletal, nervous (secondary). Cells: osteoblasts, osteoclasts, nucleus pulposus cells, inflammatory infiltrate. Lateralization: typically central/bilateral at the disc, abscesses may be asymmetric.

8. Temporal Development

Onset subacute-to-chronic and insidious; symptoms often present for weeks before diagnosis (pediatric mean delay 42 days). Course: progressive if untreated; treatment-induced remission is the norm. Critical window: early diagnosis (MRI accuracy improves after 2 weeks) and timely surgery (≤14 days) improve outcomes (Findings 7, 12).

9. Inheritance and Population

Incidence ~3/100,000/yr; elderly, male-predominant, with a toddler subgroup (Findings 2, 3). **No inheritance pattern** — acquired infection. Geographic variation reflects MRSA prevalence and IVDU epidemiology.

10. Diagnostics

Contrast MRI (gold standard) + 18F-FDG PET/CT (adjunct) + blood culture/image-guided biopsy (Finding 7). Labs: elevated CRP/ESR, WBC; positive blood cultures. Open sampling improves yield over percutaneous (PMID: 41519476). No genetic testing indicated. Differential: tuberculous spondylodiscitis, Brucella, fungal, degenerative Modic changes, malignancy.

11. Outcome/Prognosis

Mortality 3–12%; relapse 6–17%; residual disability 20–30%; 1-yr OS 76% in surgical staphylococcal cohorts (Finding 8). Prognostic factors: age, sepsis severity, prior spinal surgery, timely surgery, concurrent infection.

12. Treatment

Pathogen-directed antibiotics (6 weeks non-inferior to 12; Finding 5), with anti-staphylococcal β-lactams for MSSA and vancomycin/daptomycin/linezolid for MRSA; rifampicin for biofilm/implant disease. Surgery for defined indications (Finding 12). Salvage: dalbavancin, bacteriophage therapy (Finding 11). **NCIT suggestions:** NCIT:C254 (Antibiotic), NCIT:C1338 (Vancomycin), NCIT:C61735 (Linezolid), NCIT:C15329 (Surgery), NCIT:C157866 (Daptomycin).

13. Prevention

Primary: aseptic surgical technique, treat bacteremia sources, screen/treat asymptomatic bacteriuria before urologic procedures, harm-reduction for IVDU. Secondary: early imaging/culture in at-risk patients with back pain and fever. Tertiary: adequate antibiotic duration and timely surgery to prevent relapse/neurologic sequelae. No vaccine exists for *S. aureus*.

14. Other Species / Natural Disease

Staphylococcal discitis/spondylitis occurs naturally in veterinary species (e.g., pigs, poultry, dogs). Experimental disease is best characterized in the **rabbit** (*Oryctolagus cuniculus*, NCBI:txid9986; Finding 9). Zoonotic potential exists via livestock-associated MRSA. Orthologous host genes (e.g., TNFRSF1A, TLR2, TNFSF11/RANKL) are conserved across mammals.

15. Model Organisms

The **New Zealand White rabbit** percutaneous-inoculation model is the established in vivo system (Finding 9), reproducing disc-space narrowing, endplate erosion, epidural abscess, and paraplegia, and validated by MRI. Limitations: does not model chronic hematogenous seeding or the aged/comorbid human host. Applications: pathogenesis, antibiotic penetration, imaging validation, adjunct therapies (e.g., intradiscal vancomycin + diode laser, PMID: 22554049).

Evidence Base

PMID	Title (abbrev.)	Role
41023355	*Infectious spondylodiscitis: 14-year retrospective*	Core epidemiology, microbiology, complications (F1, F2, F8)
39716636	*Seasonal trends, Japan nationwide (n=71,134)*	Large-scale demographics, MRSA share (F1, F2, F8)
34091001	*Clinical characteristics of PVO*	*S. aureus* share, relapse/mortality, ITR predictors (F1, F4, F8)
40653018	*MRSA spinal infection outcomes*	MRSA-specific risk factors (F4)
25468170	*6 vs 12 weeks antibiotics RCT*	Landmark duration trial (F5)
26846887	*TNFR1/TLR2 in S. aureus infection*	Core mechanism (F6)
25186431	*PET/CT vs MRI prospective*	Diagnostic accuracy (F7)
42232200	*18F-FDG PET/CT in spinal infection*	Combined-modality sensitivity (F7)
27317050	*MRI vs imaging value*	MRI timing dependence (F7)
11961447	*Nontuberculous spondylodiscitis in children*	Pediatric microbiology, nosology (F3, F13)
18485785	*Rabbit disc-space infection model*	Animal model (F9)
24043368	*Antibiotic penetration in rabbit discitis*	Model application (F9)
26925420	*P. acnes vs S. aureus rabbit study*	Model comparative pathology (F9)
41946712	*African GWAS iron loci*	Polygenic host susceptibility (F10)
41478341	*Systemic phage therapy MRSA spondylodiscitis*	Salvage therapy (F11)
41305347	*Dalbavancin for bone/joint infection*	Simplified therapy (F11)
40762806	*Predictors of surgery*	Surgical indications (F12)
42314843	*Timely surgery in Staphylococcus spondylodiscitis*	Surgical timing, survival (F8, F12)
40204566	*Pyogenic Vertebral Osteomyelitis review*	Nosology, definition (F13)

PMID	Title (abbrev.)	Role
26229122 / 26316526	*IDSA 2015 NVO guidelines*	Treatment framework (F12)
39852953	*AO Spine recommendations*	Evidence synthesis, early-surgery recommendation
42262500	*Culture-negative meta-analysis*	Recurrence/disability estimates (F8)
37148755	*Concurrent infection cohort (n=10,695)*	Mortality with concurrent infection

Evidence source types: Human clinical cohorts/RCT/guidelines (F1–F5, F7, F8, F11–F13); in vitro molecular biology (F6); model organism/rabbit (F9); GWAS/computational (F10).

Limitations and Knowledge Gaps

1. **Template mismatch.** Roughly a third of the requested template (causal genes, pathogenic variants, inheritance, penetrance, expressivity, anticipation, founder effects, carrier frequency) is **not applicable** — staphylococcal discitis is an acquired infection, not a Mendelian disease. Host genetic contribution is polygenic and non-specific.
2. **Mechanism evidence is largely in vitro.** The protein A → TNFR-1 → JNK/TLR2 → RANKL cascade (F6) derives from cell-culture experiments; its quantitative contribution to human disc destruction in vivo is inferred, not directly demonstrated in patients.
3. **Discitis-specific vs spectrum-level data.** Much evidence pools discitis with vertebral osteomyelitis and epidural abscess; few studies isolate disc-only staphylococcal disease, so some frequencies (e.g., complication rates) are spectrum-level.
4. **Heterogeneous cohorts.** Retrospective designs, varying MRSA prevalence, and differing culture practices (culture-negative ~43% of cases) introduce confounding into epidemiologic and prognostic estimates.
5. **Sparse pediatric mechanistic data.** The toddler subgroup is described clinically but not mechanistically.
6. **Emerging therapies lack RCTs.** Dalbavancin and phage evidence is observational/case-based (F11), not randomized.

Proposed Follow-up Experiments / Actions

1. **In vivo validation of the RANKL axis.** Test RANKL/OPG modulation (e.g., anti-RANKL, JNK inhibitors) in the rabbit *S. aureus* discitis model to quantify effect on endplate destruction — bridging the in vitro→in vivo gap in F6.
2. **Discitis-specific registry analysis.** Extract disc-predominant (vs bone-predominant) cases from national databases to derive discitis-specific incidence, complication, and relapse rates.
3. **Host-susceptibility GWAS in spondylodiscitis.** Extend iron-regulation/immune-pathway host genetics (F10) to a spondylodiscitis-specific cohort to identify susceptibility loci beyond bacteremia.
4. **Prospective trial of shortened/oral therapy in staphylococcal subset.** Confirm the 6-week non-inferiority finding (F5) specifically in *S. aureus* discitis, including early oral switch.
5. **Controlled evaluation of salvage therapeutics.** Prospective cohorts/registries for dalbavancin and bacteriophage therapy in refractory MRSA spinal infection (F11).
6. **Biomarker-guided surgical timing.** Prospectively validate predictors (epidural/psoas abscess, Gram-positive cocci, thoracic level) and the ≤14-day surgical window (F12) to build a decision algorithm.

Consensus Answer

Staphylococcus discitis is an acquired bacterial infection of the intervertebral disc and adjacent vertebral endplates — part of the spondylodiscitis/vertebral-osteomyelitis spectrum — caused predominantly by *Staphylococcus aureus* (MSSA/MRSA) reaching the spine hematogenously, by direct inoculation, or by contiguous spread; it chiefly affects older adults (with a toddler subgroup), presents with back pain, fever, and elevated CRP/ESR, and is diagnosed by contrast MRI plus blood/biopsy culture. Bone destruction is driven by *S. aureus*–induced osteoblast/immune signaling (protein A–TNFR-1 → JNK/TLR2 → RANKL) and osteoclastic resorption, and it is treated with roughly six weeks of pathogen-directed antibiotics (6 weeks non-inferior to 12 in an RCT), reserving surgery for neurologic deficit, instability, abscess, or medical failure; mortality is ~3–12%, relapse ~6–17%, and residual

disability ~20–30%. It is not a genetic/heritable disorder — there are no causal host genes, only polygenic susceptibility to invasive *S. aureus* — so OMIM/Orphanet, inheritance, and germline-variant categories are not applicable.
